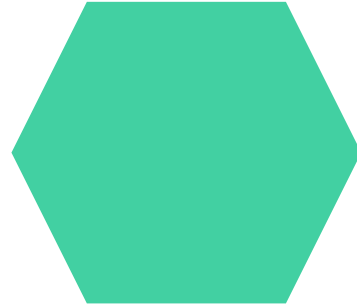
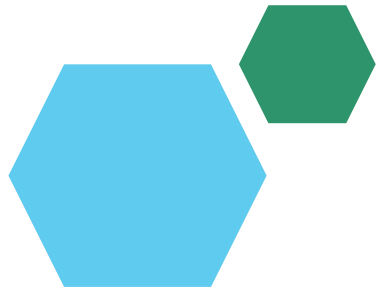




Project Title -Seasonal Agriculture Performance Analysis

Anshika Gupta

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AICTE STU ID :STU6638c7bde08f71714997181



PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

Seasonal Agriculture Performance Analysis is a data analysis project that studies agricultural performance across different seasons, crops, and regions. The project uses agricultural data related to environmental conditions, resource usage, production and economic factors. The project focuses on understanding how factors such as rainfall, soil moisture, irrigation, and farming practices can influence crop yield and overall farm performance.

The dataset contains 4,000 records, covering 8 states, 10 districts, 8 crops, 3 seasons, and 4 irrigation methods. Some values were missing in rainfall, soil moisture, and yield, so the data was cleaned before analysis.

The data is cleaned and analyzed using Python to find useful patterns, compare seasonal performance, and understand relationships between different factors and agricultural outcomes. The analysis also looks at yield, profit, resource efficiency, and disease/pest risk to identify important differences and insights.

By comparing seasons, crops, and regions, we can identify which conditions and seasons show better performance, understand differences in yield and profit, and find areas where farming practices or resource usage can be improved.

WHO ARE THE END USERS?

- Farmers
- Agriculture Departments
- Agriculture businesses
- Agriculture Researchers
- Government

Technology Used

- **Python:** Main programming language for data analysis
- **Pandas:** Data loading, cleaning, preprocessing, grouping, aggregation, and analysis
- **Numpy:** Numerical calculations and data manipulation
- **Matplotlib:** Creating basic charts and visualizations
- **Seaborn:** Creating statistical and advanced visualizations such as heatmaps, box plots, and correlation
- **Google Colab:** Development environment for performing and documenting the analysis
- **CSV:** Dataset storage



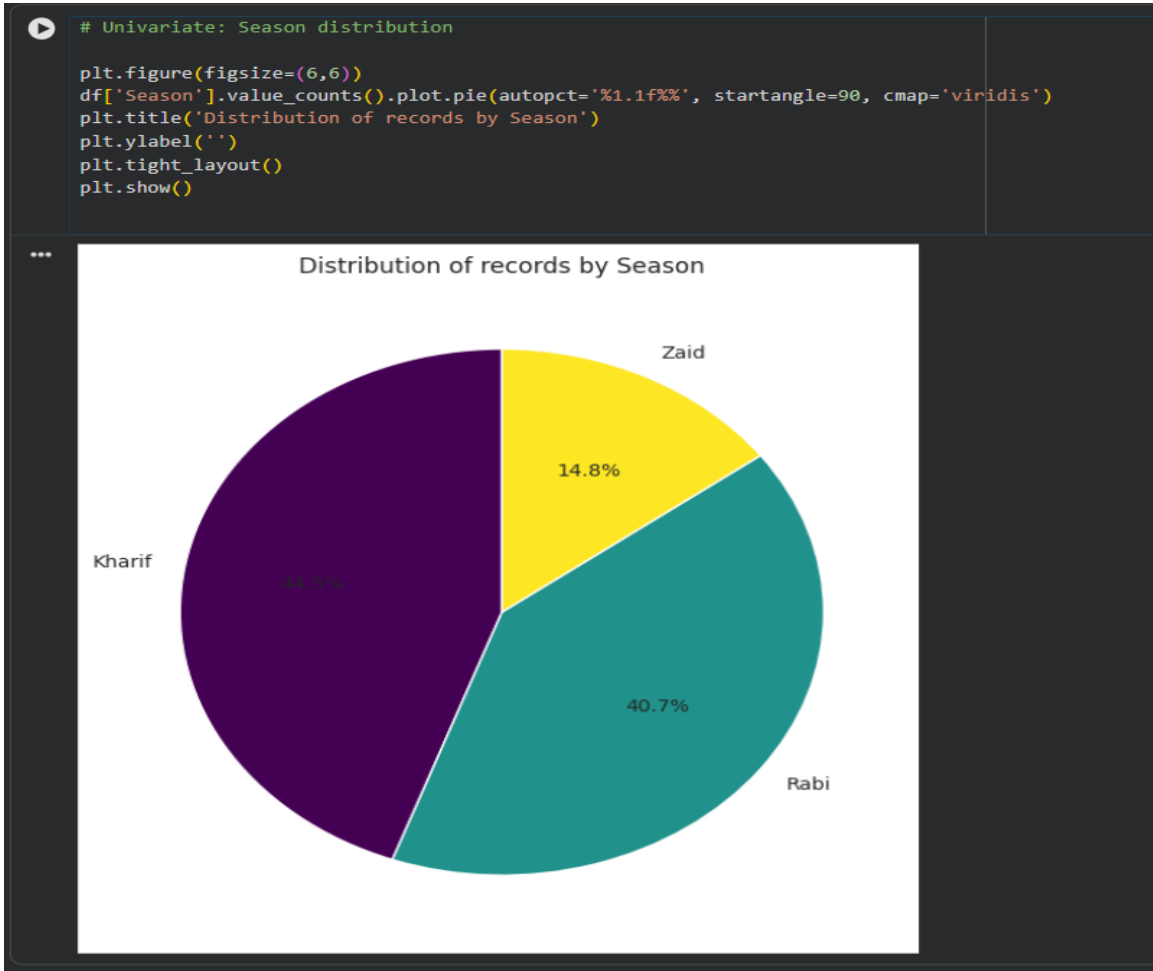
RESULTS

Descriptive statistics

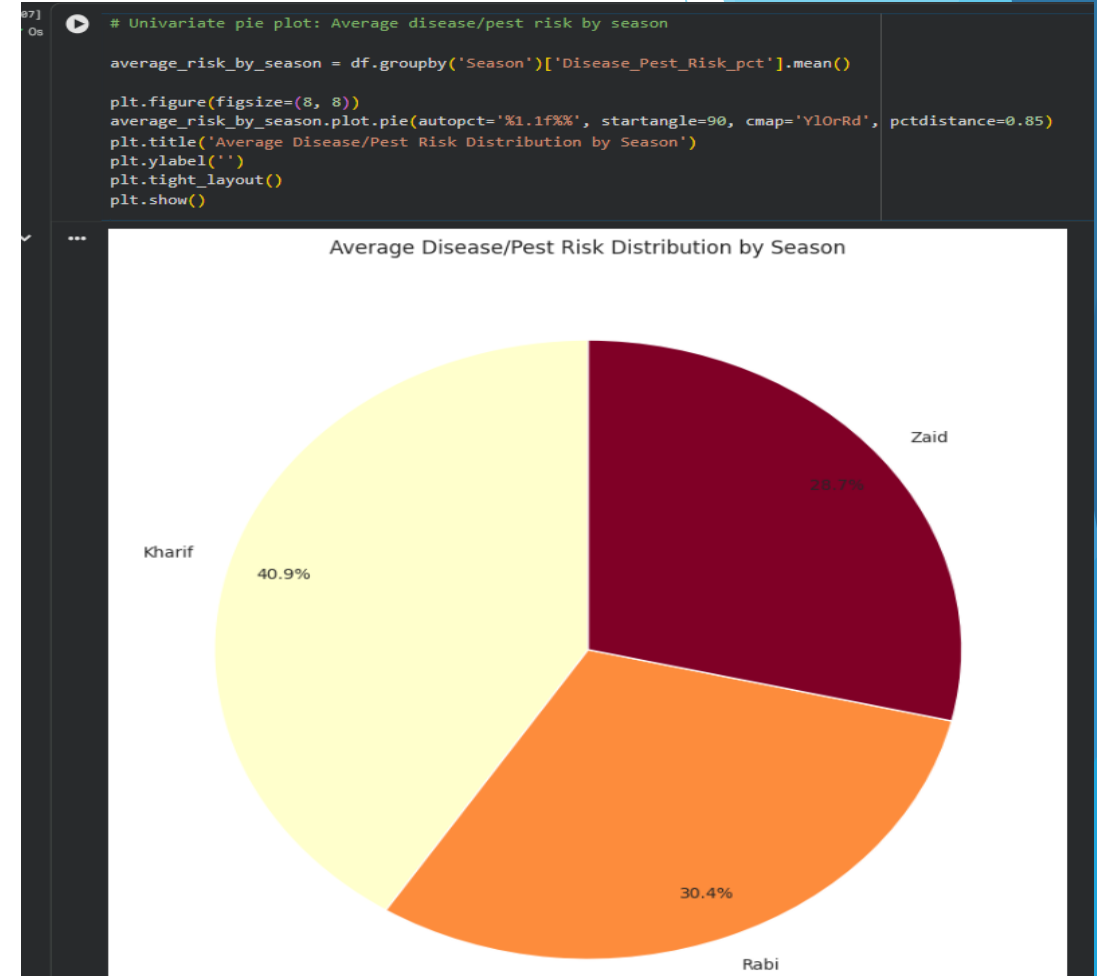
```
stats= df[numeric_col].describe().T
stats['range']=stats['max']-stats['min']
stats['IQR']=stats['75%']-stats['25%']
stats
```

	count	mean	std	min	25%	50%	75%	max	range	IQR
Farm_Area_Hectares	4,000.00	7.89	4.22	0.50	4.26	7.89	11.65	15.00	14.50	7.39
Rainfall_mm	4,000.00	600.92	287.19	80.00	371.58	582.00	831.98	1,395.20	1,315.20	460.40
Avg_Temperature_C	4,000.00	26.82	3.82	16.20	23.90	26.90	29.60	39.70	23.50	5.70
Humidity_pct	4,000.00	63.21	11.96	25.00	54.77	63.00	71.90	95.00	70.00	17.13
Sunlight_Hours_Day	4,000.00	7.32	1.12	3.50	6.60	7.30	8.10	11.00	7.50	1.50
Soil_pH	4,000.00	6.70	0.64	5.20	6.26	6.69	7.14	8.40	3.20	0.88
Soil_Moisture_pct	4,000.00	26.51	6.68	8.00	21.80	26.40	31.20	47.00	39.00	9.40
Nitrogen_kg_ha	4,000.00	120.25	31.22	40.00	98.50	120.10	141.22	220.00	180.00	42.72
Phosphorus_kg_ha	4,000.00	57.78	17.02	15.00	46.20	57.50	69.40	120.00	105.00	23.20
Potassium_kg_ha	4,000.00	104.76	27.74	30.00	86.20	105.10	123.60	200.00	170.00	37.40
Fertilizer_kg_ha	4,000.00	186.04	60.26	40.00	143.97	186.25	227.20	400.00	360.00	83.22
Pesticide_Litre_ha	4,000.00	5.07	1.98	0.50	3.69	5.12	6.40	12.08	11.58	2.71
Seed_Quality_Score	4,000.00	0.83	0.10	0.65	0.74	0.83	0.91	1.00	0.35	0.17
Yield_Tonnes_Ha	4,000.00	5.24	13.31	0.30	1.05	1.74	2.83	101.44	101.14	1.78
Production_Tonnes	4,000.00	43.25	126.88	0.15	5.02	11.77	24.36	1,424.40	1,424.25	19.34
Market_Price_INR_Tonne	4,000.00	44,714.17	30,266.16	2,594.00	21,429.25	25,913.00	68,279.00	131,240.00	128,646.00	46,849.75
Total_Cost_INR	4,000.00	526,303.58	294,419.14	26,826.00	274,577.50	519,082.50	761,996.50	1,349,990.00	1,323,164.00	487,419.00
Revenue_INR	4,000.00	637,860.05	636,974.71	6,107.00	207,517.25	456,830.50	836,874.75	5,080,900.00	5,074,793.00	629,357.50
Profit_INR	4,000.00	111,556.46	545,081.89	-1,019,223.00	-148,731.75	3,673.00	216,187.75	4,345,421.00	5,364,644.00	364,919.50
Water_Used_m3	4,000.00	6,045.57	5,572.94	128.00	2,268.75	4,435.00	7,829.75	40,902.00	40,774.00	5,561.00
Water_Efficiency_t_per_1000m3	4,000.00	5.39	8.84	0.14	1.66	2.94	5.15	79.80	79.66	3.49
Disease_Pest_Risk_pct	4,000.00	46.37	12.42	5.00	37.30	46.40	54.90	88.90	83.90	17.60

RESULTS



- This pie chart shows percentage of records distribution by Season.
- Farming in Kharif and Rabi Season is high and Zaid is low

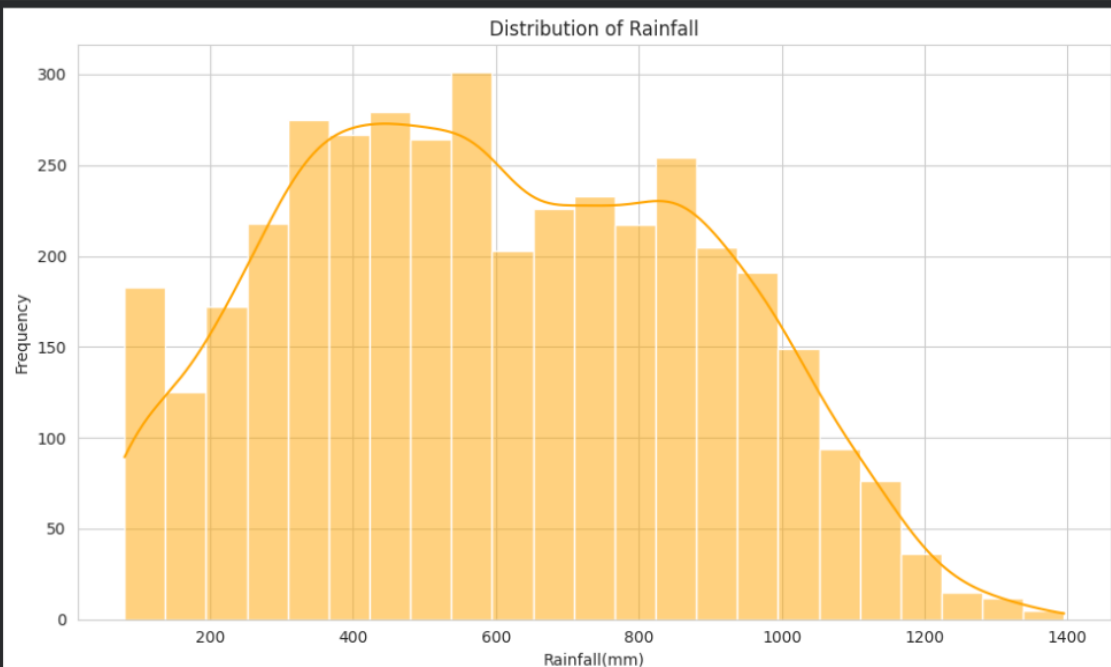


- This pie chart shows average risk of disease in each season.
- Kharif season has high risk of pest so we have to take care more on this season.

RESULTS

Univariate: Rainfall distribution

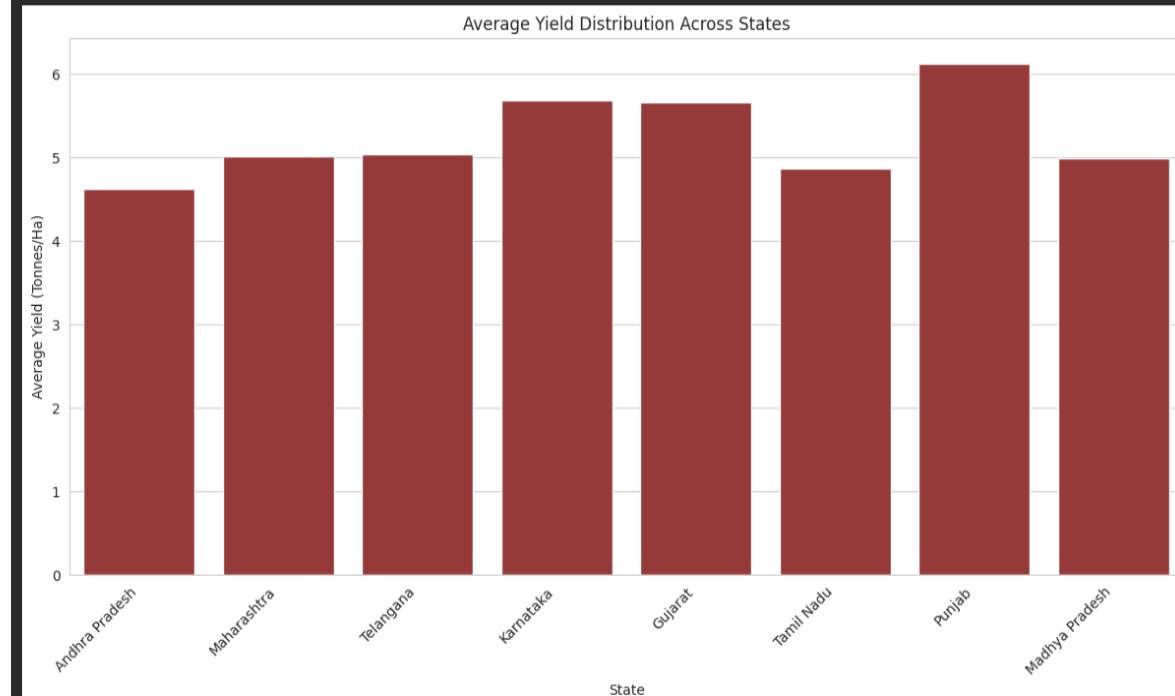
```
plt.figure(figsize=(10,6))
sns.histplot(data=df, x='Rainfall_mm', kde=True, color='orange')
plt.title('Distribution of Rainfall')
plt.xlabel('Rainfall(mm)')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



- Rainfall values are spread across a wide range, showing different rainfall conditions in the dataset.
- Most observations are concentrated around the middle range, while very high and very low rainfall values occur less frequently.

#Bivariate: Average yield distribution across state

```
plt.figure(figsize=(12, 7))
sns.barplot(data=df, x='State', y='Yield_Tonnes_Ha', errorbar=None, color='brown')
plt.title('Average Yield Distribution Across States')
plt.xlabel('State')
plt.ylabel('Average Yield (Tonnes/Ha)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

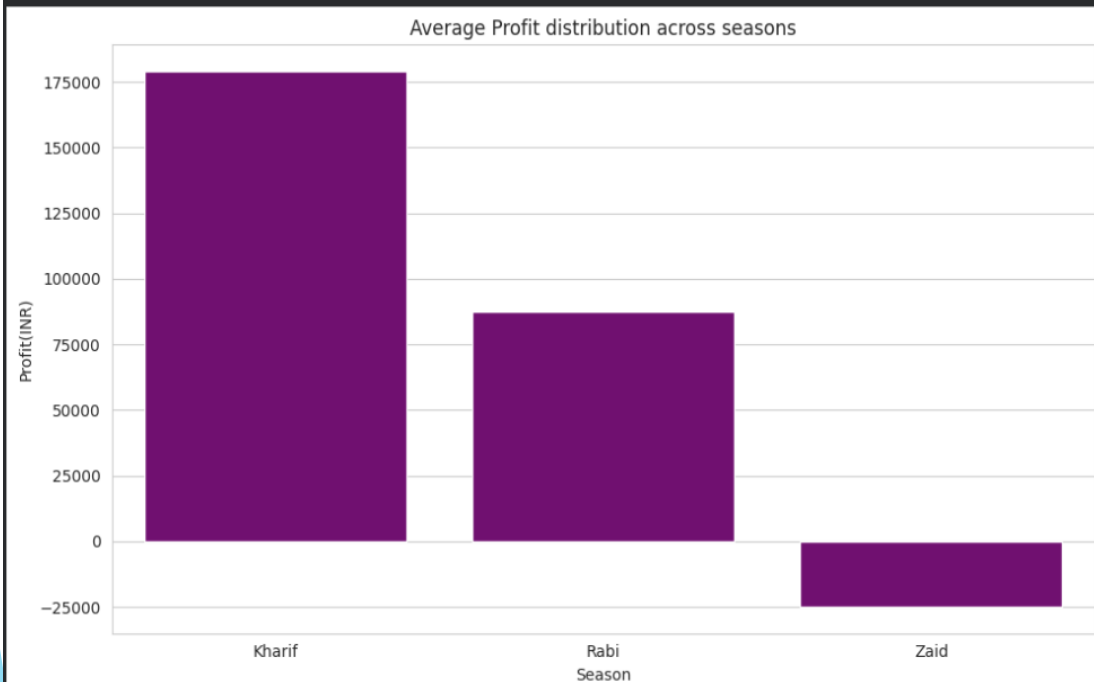


- This barplot shows average yield across states.
- This shows that agricultural performance is not the same across all regions.

RESULTS

```
#Bivariate: Seasons vs Average Profit
```

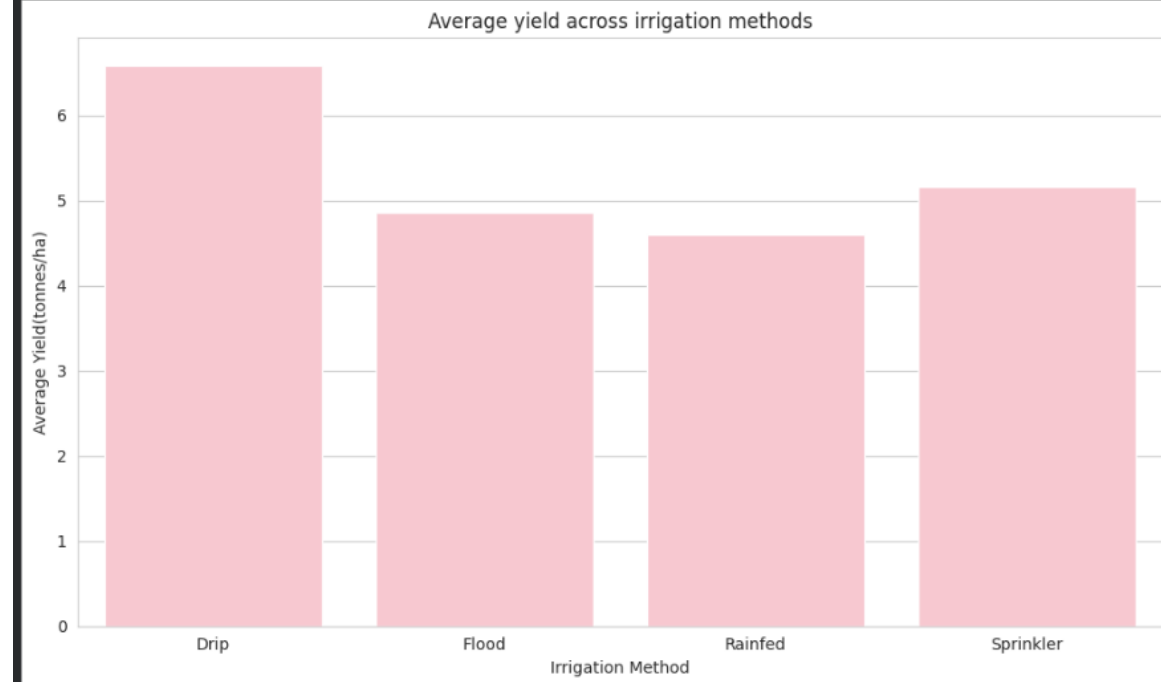
```
plt.figure(figsize=(10,6))
sns.barplot(data=df, x='Season', y='Profit_INR', errorbar=None, color='purple')
plt.title('Average Profit distribution across seasons')
plt.xlabel('Season')
plt.ylabel('Profit(INR)')
plt.tight_layout()
plt.show()
```



- This barplot shows average profit across the seasons.
- Kharif season has highest profit and Zaid season has loss.

```
# Bivariate: Irrigation method vs yield
```

```
plt.figure(figsize=(10,6))
sns.barplot(data=df, x='Irrigation_Method', y='Yield_Tonnes_Ha', errorbar=None, color='pink')
plt.title('Average yield across irrigation methods')
plt.xlabel('Irrigation Method')
plt.ylabel('Average Yield(tonnes/ha)')
plt.tight_layout()
plt.show()
```

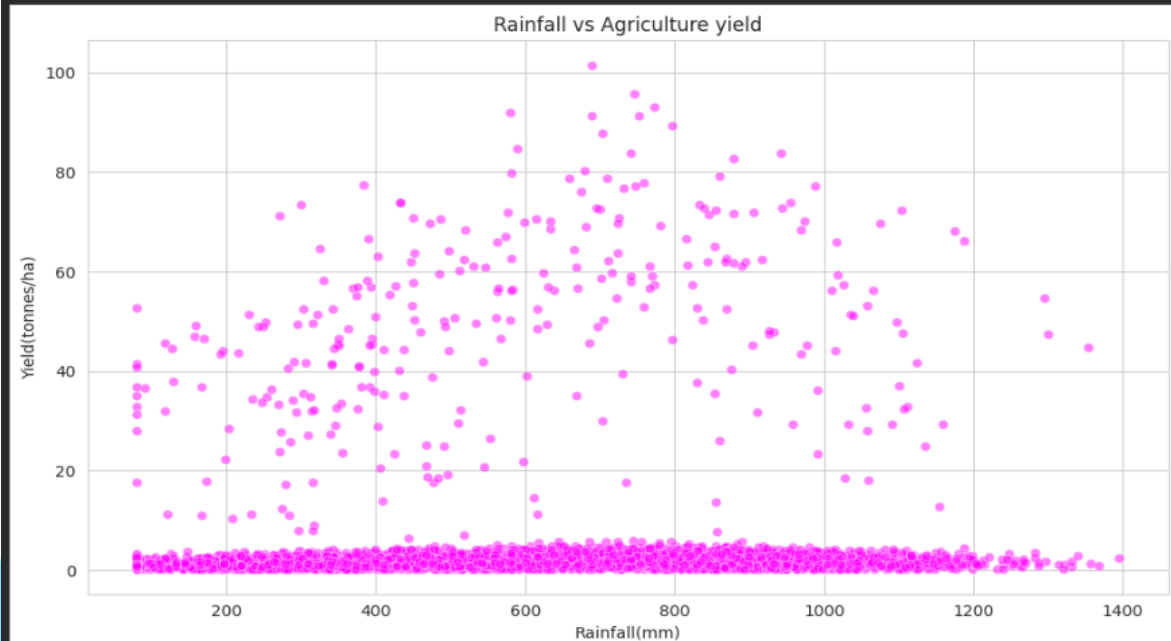


- This barplot shows average yield across the irrigation method
- Drip Irrigation method is best for high yield.

RESULTS

```
# Bivariate: Rainfall vs yield
```

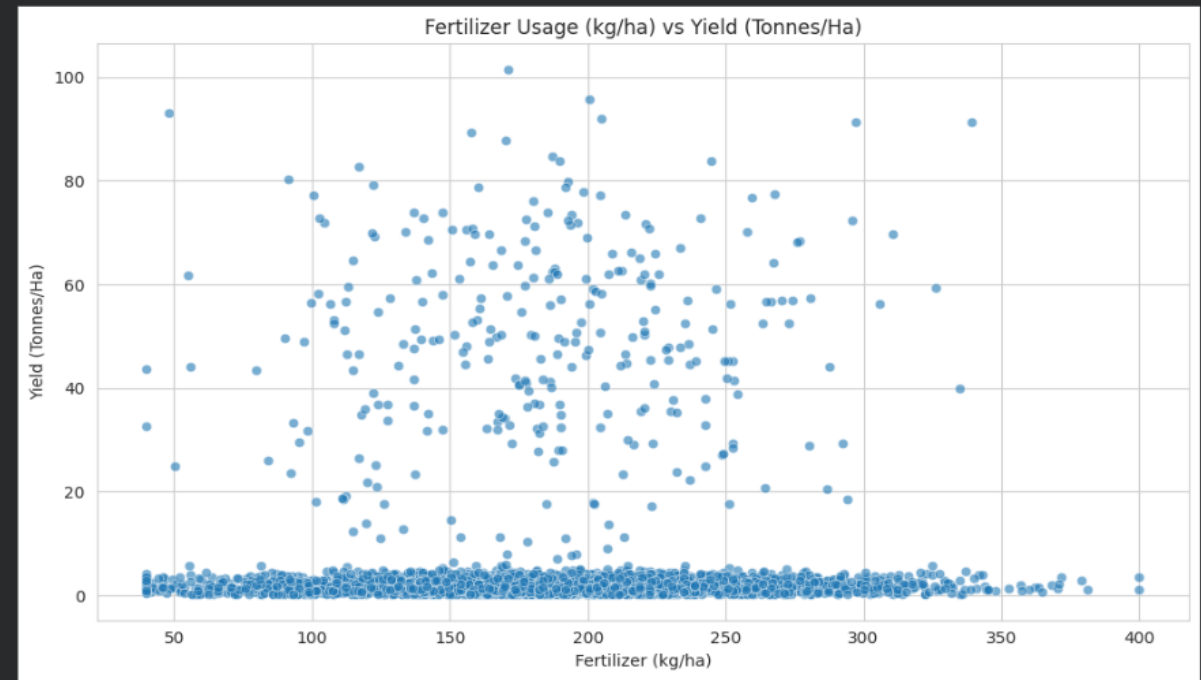
```
plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='Rainfall_mm', y='Yield_Tonnes_Ha', alpha=0.5, color='magenta')
plt.title('Rainfall vs Agriculture yield')
plt.xlabel('Rainfall(mm)')
plt.ylabel('Yield(tonnes/ha)')
plt.tight_layout()
plt.show()
```



- The scatter plot shows a very weak relationship between rainfall and crop yield.
- Most points are widely scattered, meaning higher rainfall does not consistently result in higher yield.

```
#Bivariate: Fertilizer Usage vs Agriculture Performance
```

```
plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='Fertilizer_kg_ha', y='Yield_Tonnes_Ha', alpha=0.6)
plt.title('Fertilizer Usage (kg/ha) vs Yield (Tonnes/Ha)')
plt.xlabel('Fertilizer (kg/ha)')
plt.ylabel('Yield (Tonnes/Ha)')
plt.tight_layout()
plt.show()
```

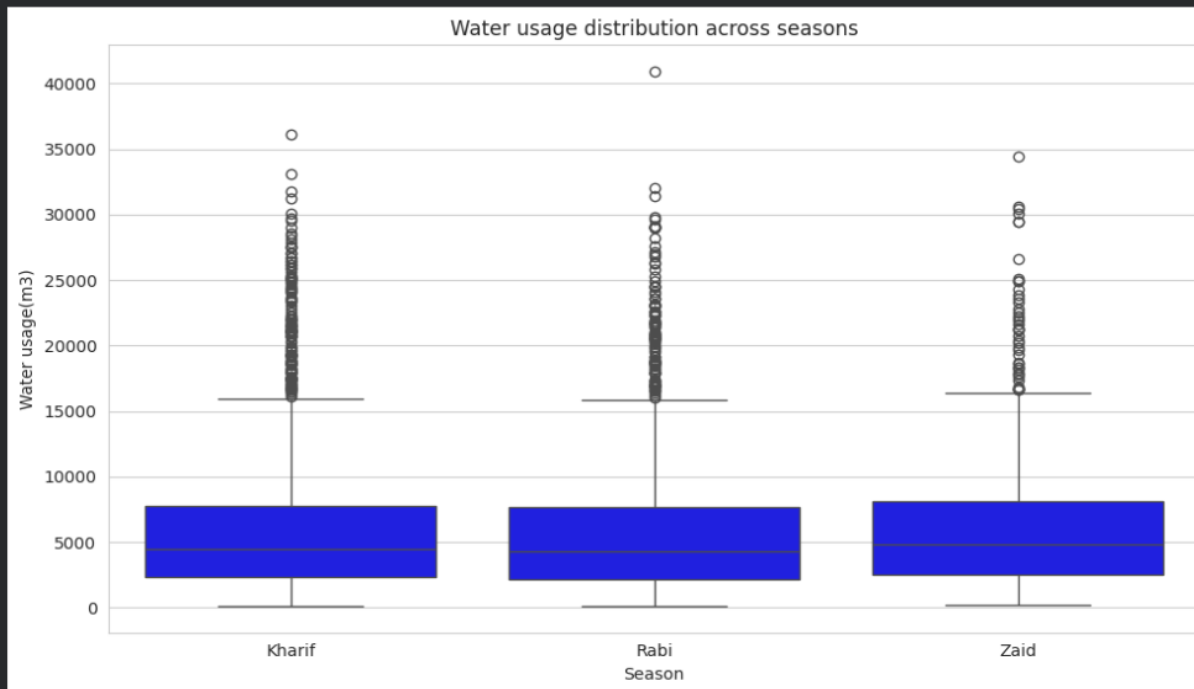


- The scatter plot shows a weak relationship between fertilizer usage and crop yield.
- Increasing fertilizer usage does not consistently lead to higher yield, as the points are widely scattered.

RESULTS

```
#Bivariate: Season vs Water usage
```

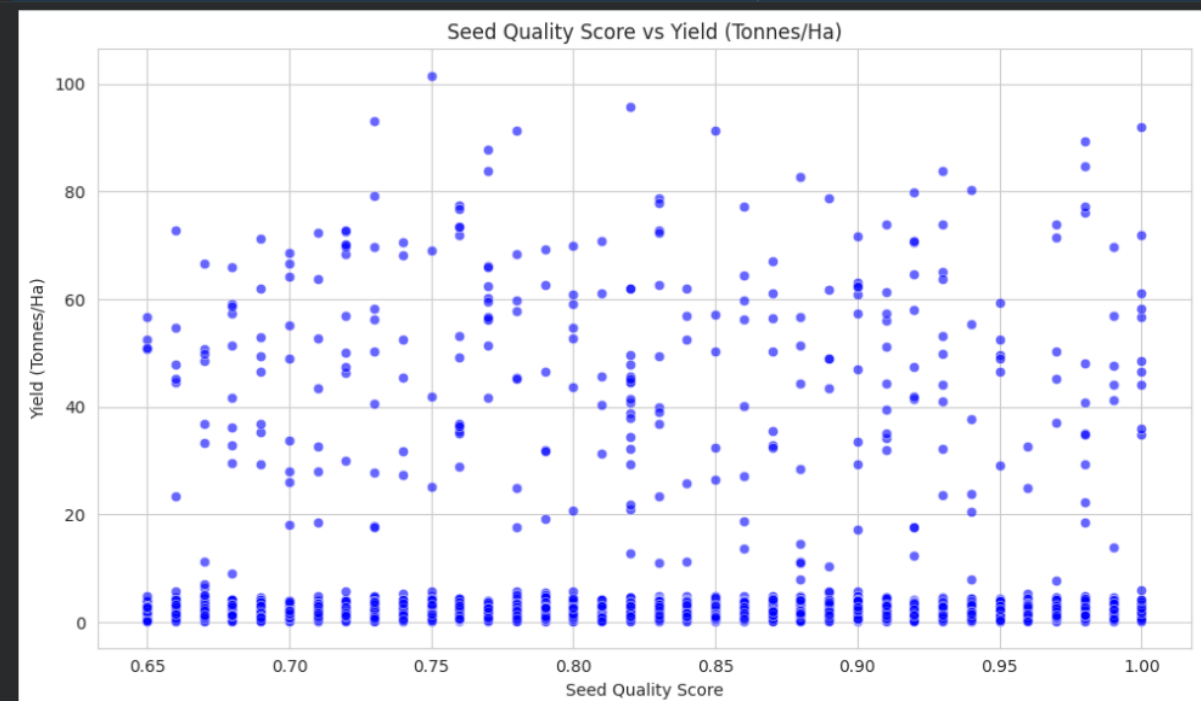
```
plt.figure(figsize=(10,6))
sns.boxplot(data=df, x='Season', y='Water_Used_m3', color='blue')
plt.title('Water usage distribution across seasons')
plt.xlabel('Season')
plt.ylabel('Water usage(m3)')
plt.tight_layout()
plt.show()
```



- Water usage shows variation across all three seasons.
- The boxplots show that water consumption has a wide spread within each season, meaning farms use different amounts of water.

```
# Bivariate: Seed Quality vs Yield
```

```
plt.figure(figsize=(10,6))
sns.scatterplot(data=df, x='Seed_Quality_Score', y='Yield_Tonnes_Ha', alpha=0.6, color='blue')
plt.title('Seed Quality Score vs Yield (Tonnes/Ha)')
plt.xlabel('Seed Quality Score')
plt.ylabel('Yield (Tonnes/Ha)')
plt.tight_layout()
plt.show()
```



- The scatter plot shows a very weak relationship between seed quality score and crop yield.
- The points are widely scattered, meaning a higher seed quality score does not consistently lead to higher yield.

RESULTS

```
# Bivariate: Average Water Efficiency by Irrigation Method
```

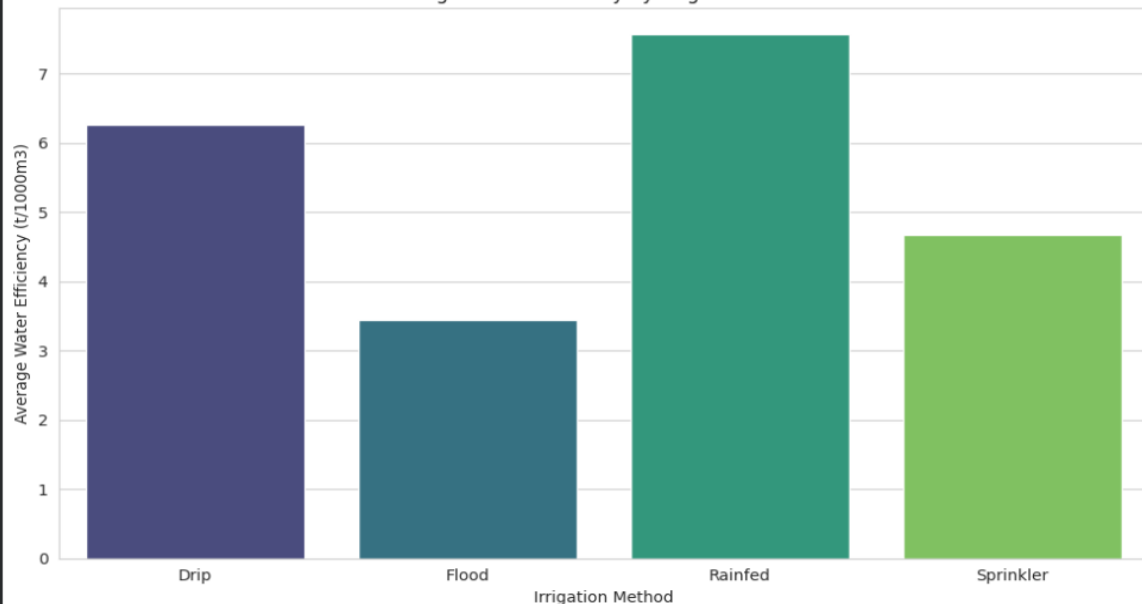
```
plt.figure(figsize=(10,6))
sns.barplot(data=df, x='Irrigation_Method', y='Water_Efficiency_t_per_1000m3', errorbar=None, palette='viridis')
plt.title('Average Water Efficiency by Irrigation Method')
plt.xlabel('Irrigation Method')
plt.ylabel('Average Water Efficiency (t/1000m3)')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_1632/1651336976.py:4: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `
```

```
sns.barplot(data=df, x='Irrigation_Method', y='Water_Efficiency_t_per_1000m3', errorbar=None, palette='viridis')
```

Average Water Efficiency by Irrigation Method



- The barplot shows average water efficiency by irrigation.
- Rainfed method is more efficient and Flood is very less efficient.

```
# Bivariate: Average Water Efficiency across Seasons
```

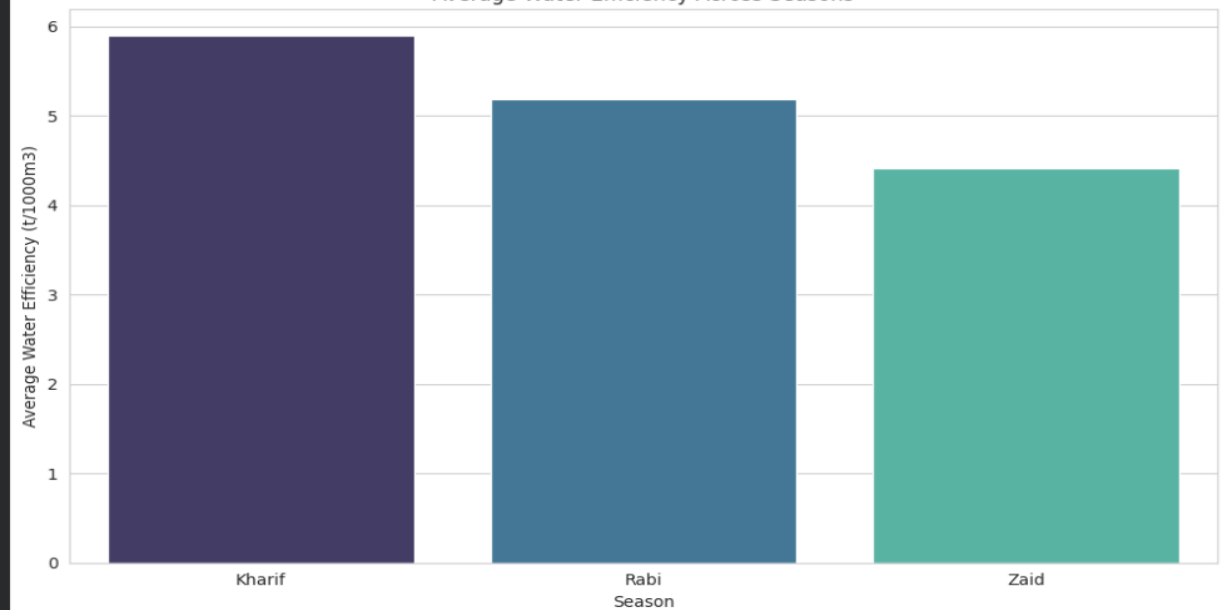
```
plt.figure(figsize=(10,6))
sns.barplot(data=df, x='Season', y='Water_Efficiency_t_per_1000m3', errorbar=None, palette='mako')
plt.title('Average Water Efficiency Across Seasons')
plt.xlabel('Season')
plt.ylabel('Average Water Efficiency (t/1000m3)')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_1632/1675740506.py:4: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `
```

```
sns.barplot(data=df, x='Season', y='Water_Efficiency_t_per_1000m3', errorbar=None, palette='mako')
```

Average Water Efficiency Across Seasons

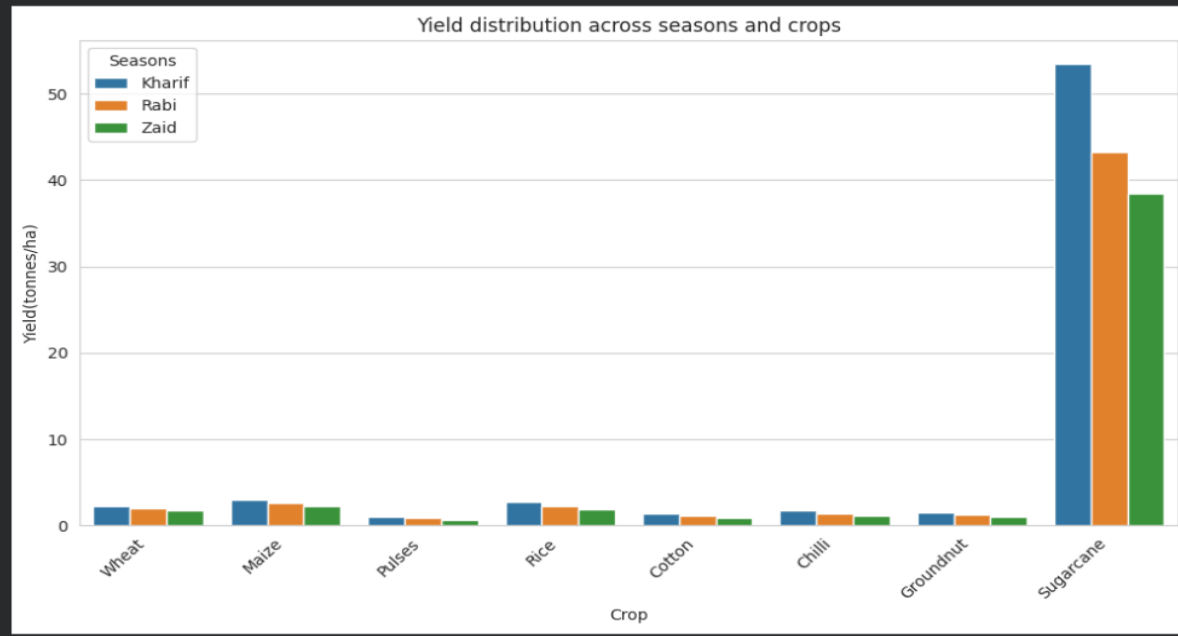


- The barplot shows average water efficiency across seasons.
- Kharif season has higher water efficiency as compare to other seasons.

RESULTS

```
# Multivariate: Seasons, crop and yield
```

```
plt.figure(figsize=(10,6))
sns.barplot(data=df, x='Crop', y='Yield_Tonnes_Ha', hue='Season', errorbar=None)
plt.xticks(rotation=45, ha='right')
plt.title('Yield distribution across seasons and crops')
plt.xlabel('Crop')
plt.ylabel('Yield(tonnes/ha)')
plt.legend(title='Seasons')
plt.tight_layout()
plt.show()
```



- This barplot shows crop yield across all the seasons.
- It clearly shows sugarcane has much higher yield than the other crops.

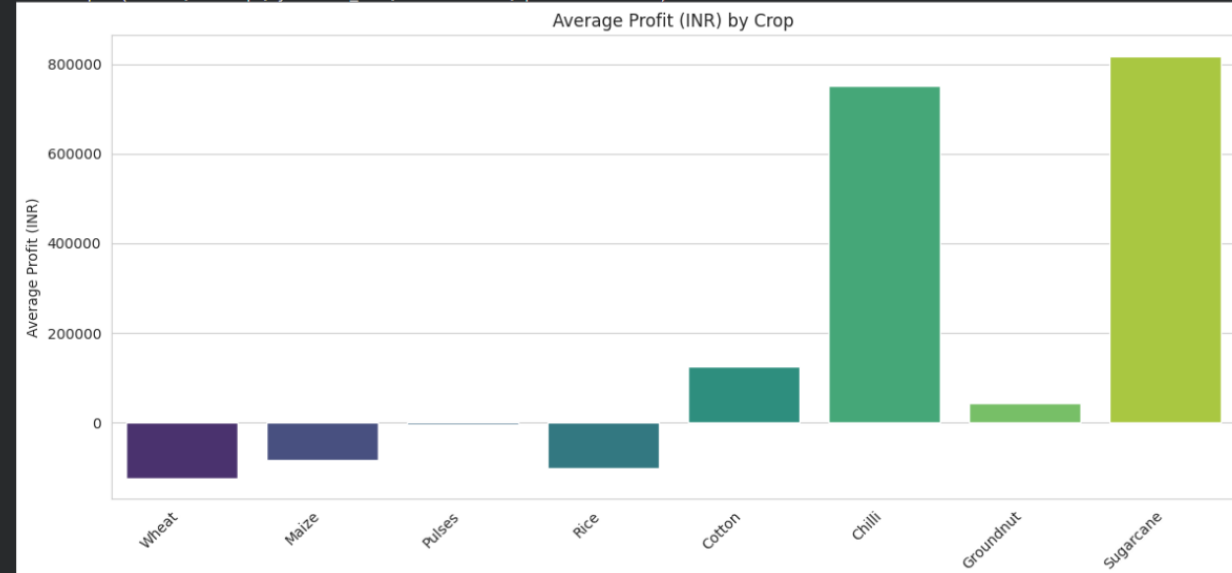
```
#Crop vs profit
```

```
plt.figure(figsize=(12, 6))
sns.barplot(data=df, x='Crop', y='Profit_INR', errorbar=None, palette='viridis')
plt.title('Average Profit (INR) by Crop')
plt.xlabel('Crop')
plt.ylabel('Average Profit (INR)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

/tmp/ipykernel_1632/3678413413.py:2: FutureWarning:

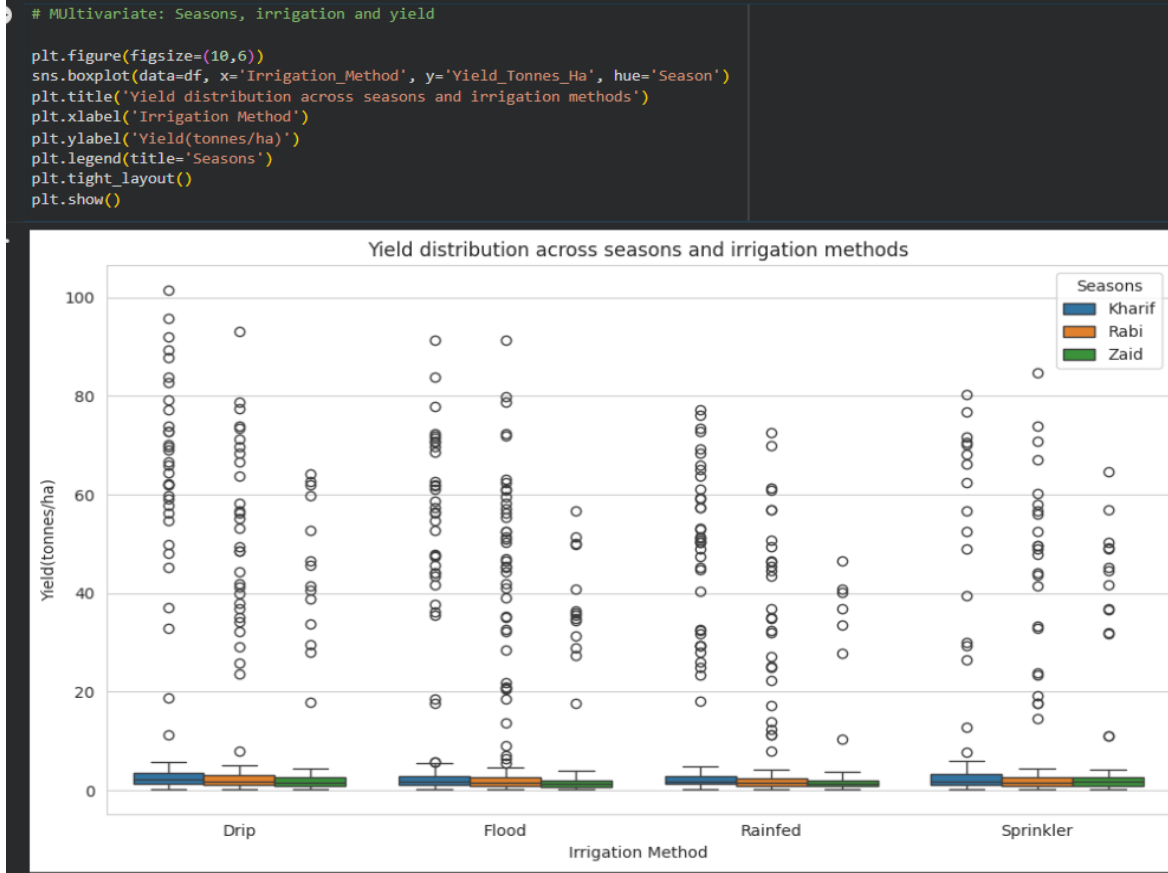
Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'x' variable to 'hue' and set 'legend=False' for the same.

```
sns.barplot(data=df, x='Crop', y='Profit_INR', errorbar=None, palette='viridis')
```

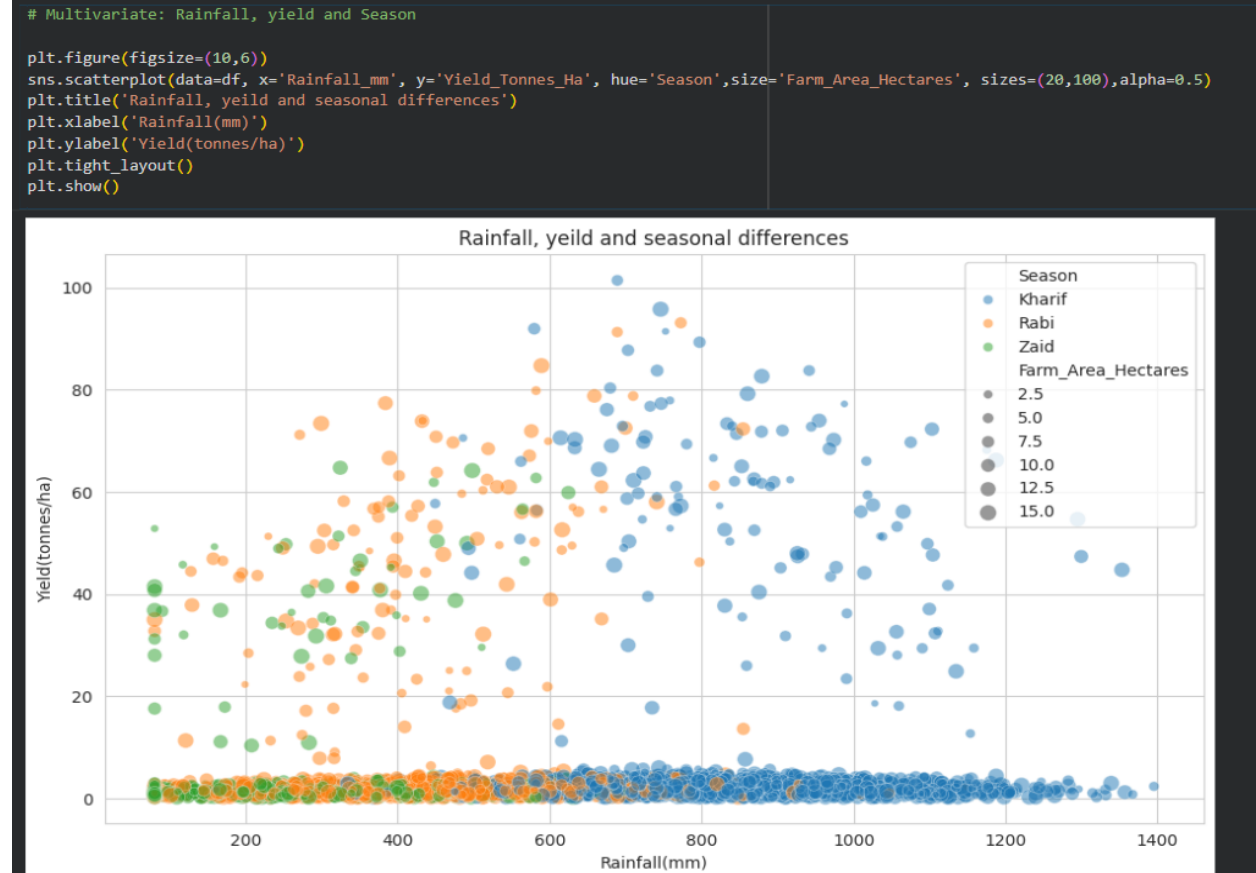


- This barplot shows average profit by crops.
- Sugarcane and chilli have highest profit than others.
- It also shows that profit varies according to yield.

RESULTS



- Kharif has the highest yield , Rabi has the median and zaid has lowest yield.
- Kharif also has the highest average yield (**5.63 tonnes/ha**), followed by Rabi (**5.04**) and Zaid (**4.64**).



- The scatter plot shows a very weak relationship between rainfall and crop yield.
- Data points for Kharif, Rabi, and Zaid overlap considerably, so there is no clear separation based only on rainfall.
- Kharif generally has higher yield values, but this cannot be explained by rainfall alone.

RESULTS

crop_summary

Season	Crop	Avg_yield	Total_prod	Avg_water_used	Avg_temp	Avg_humidity	Avg_rainfall	Avg_profit	Total_profit
Kharif	Chilli	1.73	2,675.74	4,635.06	28.47	71.02	843.14	954,380.42	176560378
	Cotton	1.37	2,237.84	5,452.45	28.58	71.25	849.47	216,999.55	46654903
	Groundnut	1.48	2,024.62	3,338.39	28.41	72.49	845.46	108,265.44	20895230
	Maize	2.97	5,463.54	4,329.20	28.67	71.94	861.22	-39,332.67	-9203845
	Pulses	1.04	1,831.36	2,853.42	28.56	71.52	835.27	43,338.90	9577896
	Rice	2.70	7,049.19	11,378.45	28.19	71.84	840.33	-64,903.40	-20379669
	Sugarcane	53.46	55,759.37	13,689.51	28.46	71.15	859.18	1,000,790.81	125098851
Rabi	Wheat	2.25	5,345.95	4,294.81	28.42	72.66	861.48	-105,871.87	-30914586
	Chilli	1.47	1,787.45	4,351.58	23.50	57.50	430.87	638,572.50	104087317
	Cotton	1.20	1,897.64	5,377.81	23.35	57.78	465.70	107,343.59	21576061
	Groundnut	1.23	1,603.90	3,490.11	23.38	59.05	409.11	10,416.54	1843728
	Maize	2.60	5,030.78	4,416.94	23.45	57.39	424.85	-89,133.40	-20768083
	Pulses	0.88	1,533.28	2,867.45	23.54	57.61	447.75	-14,309.14	-3047846
	Rice	2.32	4,719.24	10,178.32	23.38	57.93	432.06	-98,427.87	-26969236
Zaid	Sugarcane	43.29	47,118.96	13,757.14	23.62	58.58	428.50	731,612.86	94378059
	Wheat	2.06	3,807.63	3,804.25	23.73	57.74	454.56	-119,954.57	-28429232
	Chilli	1.20	638.56	4,986.30	31.70	51.79	321.07	448,659.09	28714182
	Cotton	0.96	637.62	5,385.88	30.87	53.03	295.69	-53,925.34	-4961131
	Groundnut	1.04	465.55	3,792.30	30.68	52.17	282.57	-68,872.50	-3719115
	Maize	2.29	1,507.08	4,404.45	30.92	53.04	316.84	-194,049.17	-16300130
	Pulses	0.67	306.46	2,647.19	30.94	53.65	309.96	-139,227.81	-8632124
	Rice	1.90	1,540.72	11,502.52	31.20	51.17	307.93	-227,239.34	-23178413
	Sugarcane	38.42	16,807.64	14,763.53	30.78	49.44	296.85	583,635.80	29765426
	Wheat	1.75	1,194.72	3,926.02	31.11	51.31	300.88	-193,208.95	-16422761

```
# General seasonal summary

season_summary=df.groupby('Season').agg(Records=('Farm_ID', 'count'),
                                         Average_yield=('Yield_Tonnes_Ha', 'mean'),
                                         Total_prod=('Production_Tonnes', 'sum'),
                                         Avg_water_used=('Water_Used_m3', 'mean'),
                                         Avg_temp=('Avg_Temperature_C', 'mean'),
                                         Avg_humidity=('Humidity_pct', 'mean'),
                                         Avg_rainfall=('Rainfall_mm', 'mean'),
                                         Avg_profit=('Profit_INR', 'mean'),
                                         Total_profit=('Profit_INR', 'sum'))

season_summary
```

	Records	Average_yield	Total_prod	Avg_water_used	Avg_temp	Avg_humidity	Avg_rainfall	Avg_profit	Total_profit
Season									
Kharif	1779	5.63	82,387.61	6,102.20	28.45	71.81	849.20	178,914.65	318289158
Rabi	1627	5.04	67,498.88	5,846.99	23.49	57.89	437.62	87,689.47	142670768
Zaid	594	4.64	23,098.35	6,419.89	31.04	52.01	304.65	-24,804.82	-14734066

```
state_summary = df.groupby('State').agg(
    Average_Yield=('Yield_Tonnes_Ha', 'mean'),
    Average_Profit=('Profit_INR', 'mean'),
    Average_Water_Efficiency=('Water_Efficiency_t_per_1000m3', 'mean')
).sort_values('Average_Yield', ascending=False)

display(state_summary.round(2))
```

	Average_Yield	Average_Profit	Average_Water_Efficiency
State			
Punjab	6.12	136,025.64	5.62
Karnataka	5.69	119,422.34	5.56
Gujarat	5.66	117,568.24	5.58
Telangana	5.04	108,829.62	5.46
Maharashtra	5.02	135,428.86	5.28
Madhya Pradesh	4.99	86,840.76	5.27
Tamil Nadu	4.88	117,744.75	5.31
Andhra Pradesh	4.63	73,453.53	5.03

Future scope

- **Add more data:** Include data from more years, states, districts, and crops to get a broader view of agricultural performance.
- **Weather-based analysis:** Use real-time weather data such as rainfall and temperature to improve seasonal analysis.
- **Crop recommendation:** Develop a system that suggests suitable crops based on season, soil, weather, and available resources.
- **Yield prediction:** Use machine learning models to predict crop yield before harvesting.
- **Better resource planning:** Analyze water, fertilizer, and pesticide usage to find more efficient farming practices.
- **Early risk detection:** Use the data to identify possible disease, pest, or weather-related risks at an early stage.
- **Interactive dashboard:** Build a dashboard using **Power BI or a web application** so farmers and other users can easily explore the results.

GitHub Link

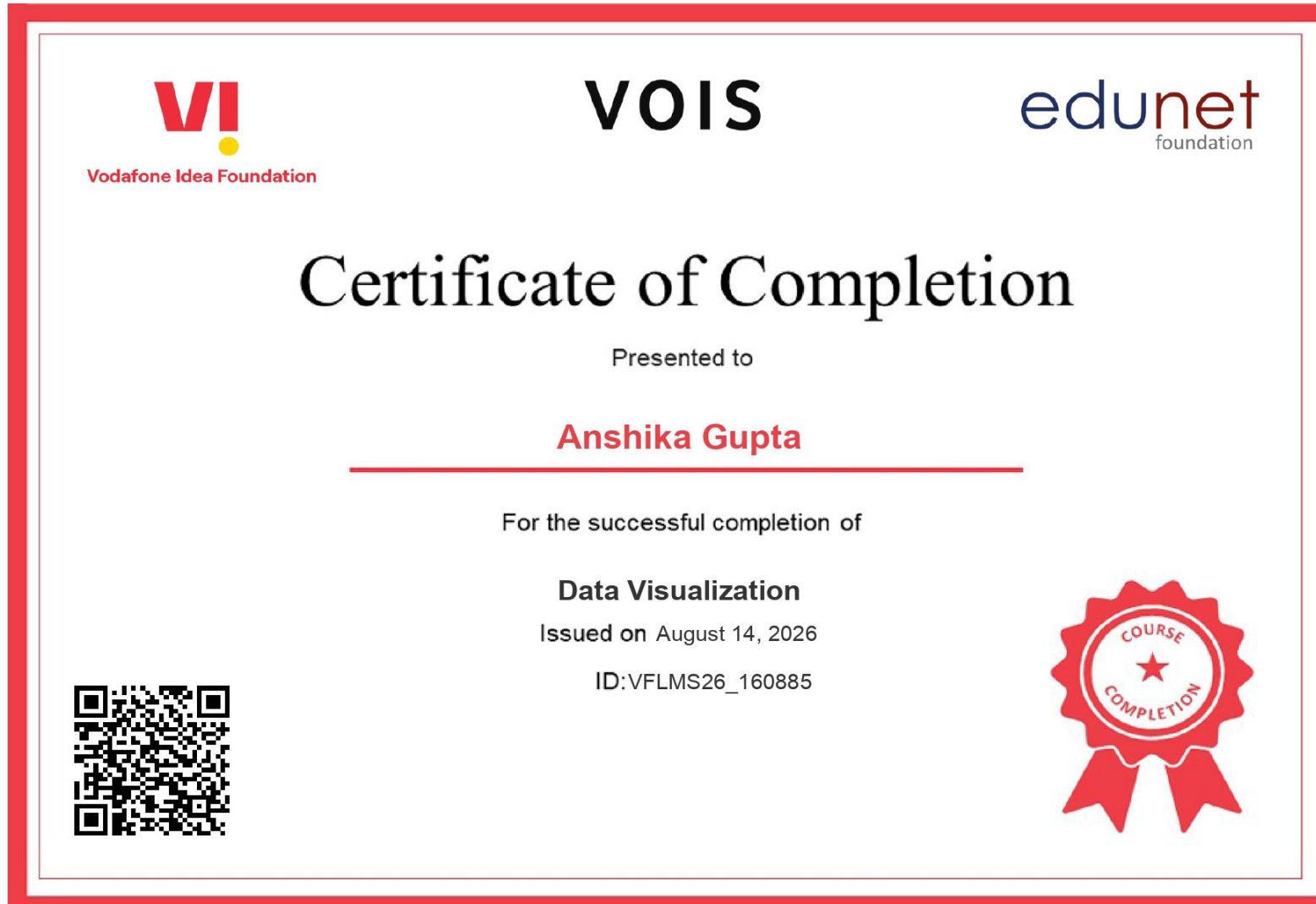
Github link

<https://github.com/Anshika1000/Seasonal-Agriculture-Performance-Analysis.git>



VOIS Course completion certificate -

Course name- Data Visualization



Thank you